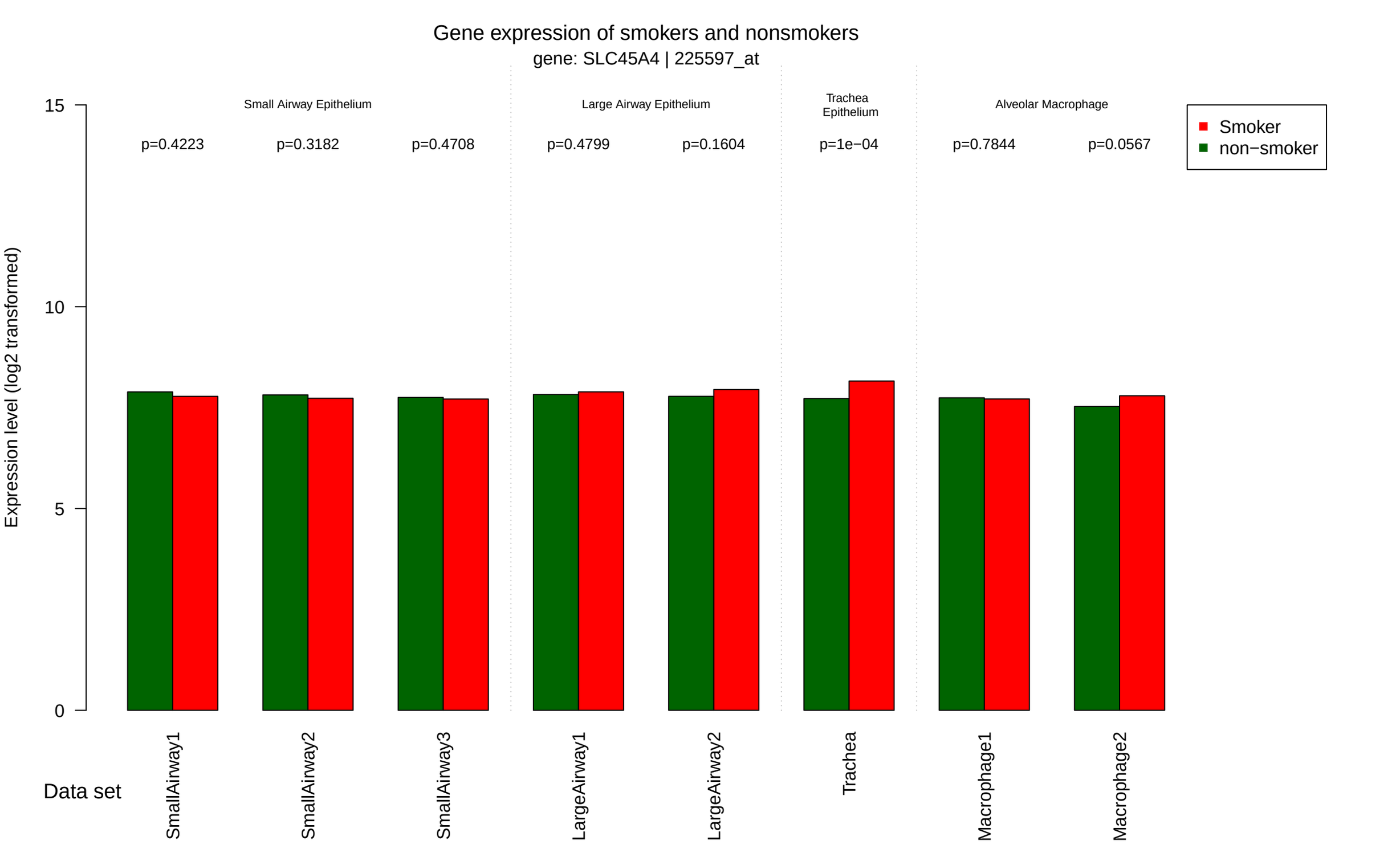
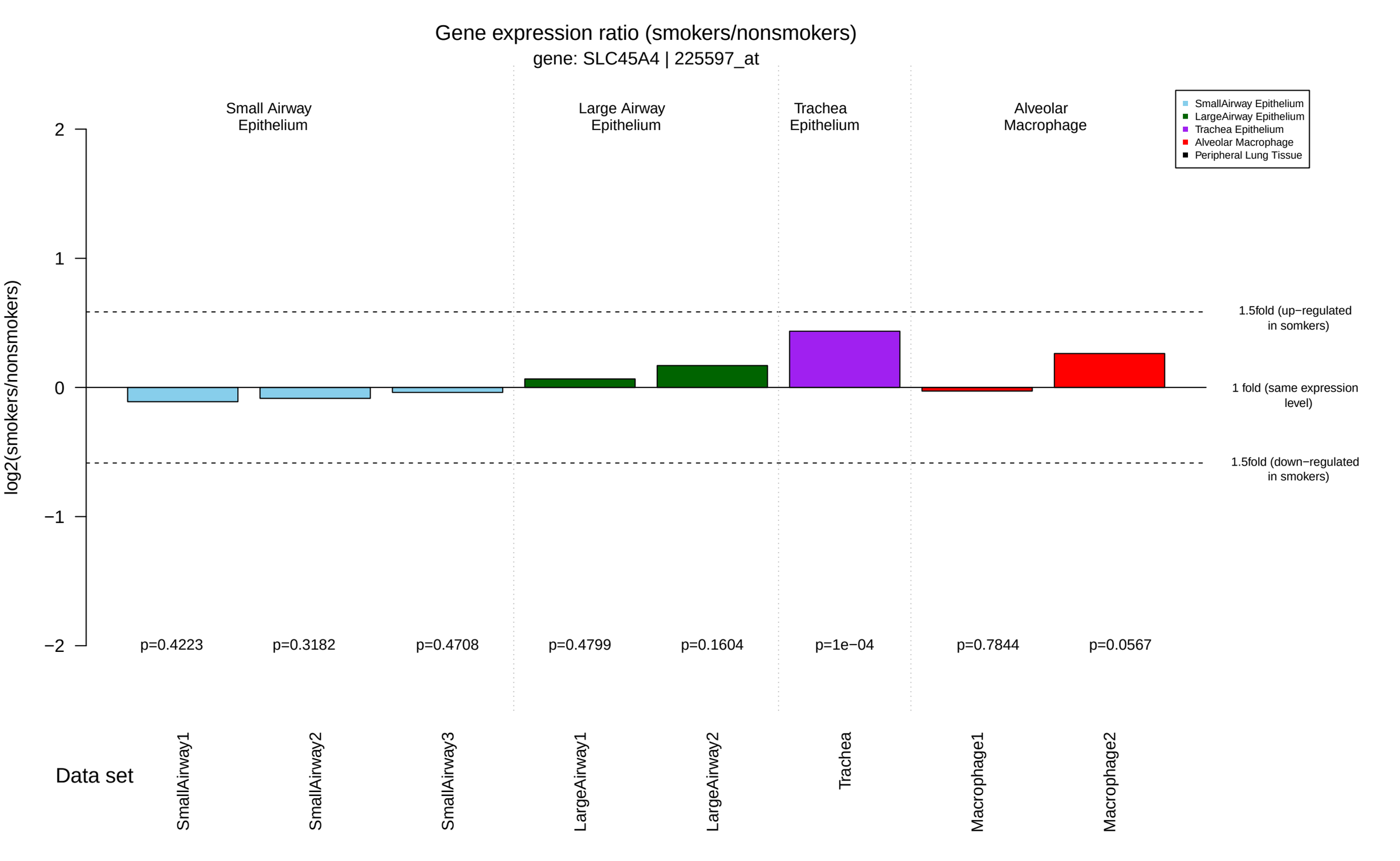
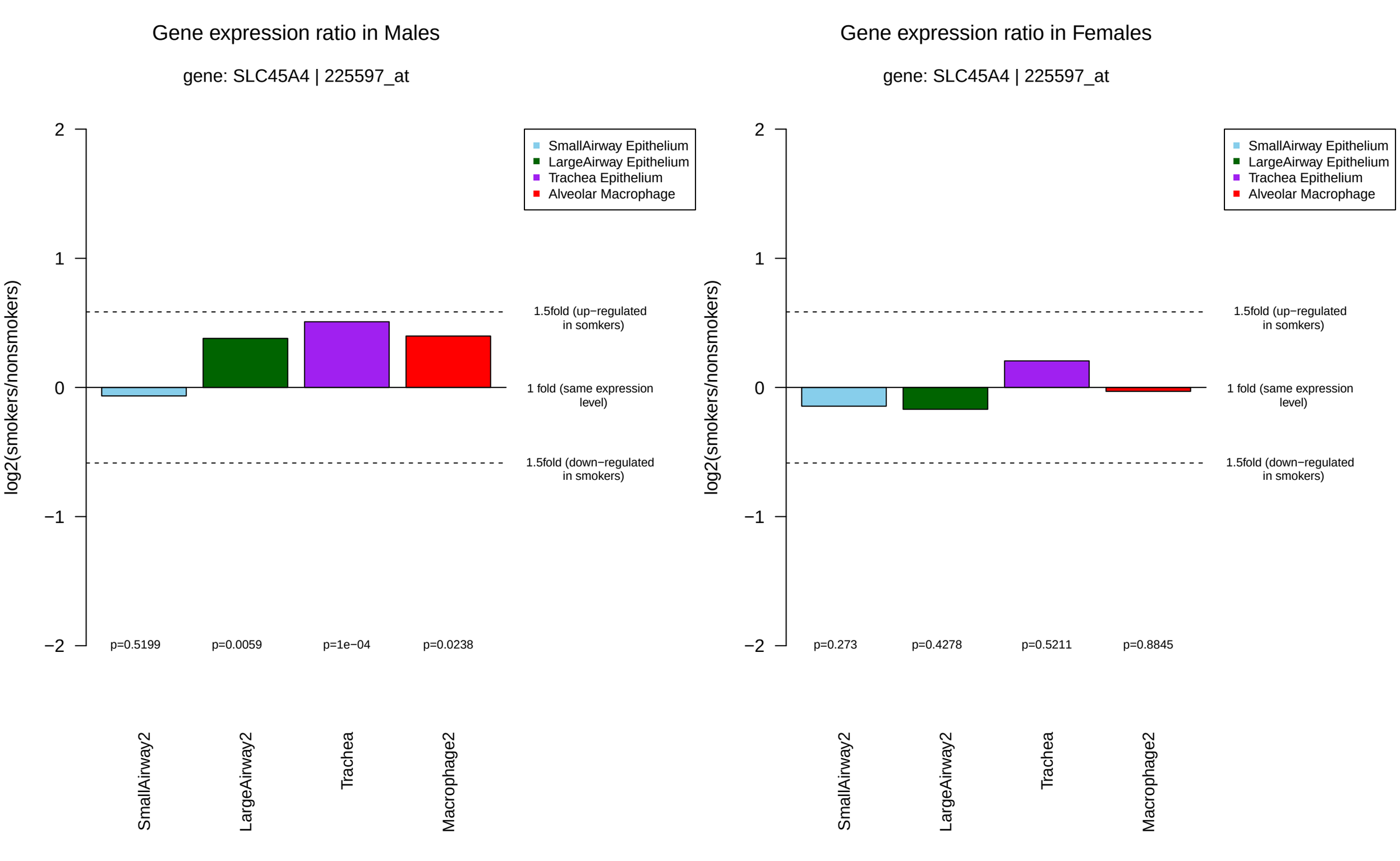
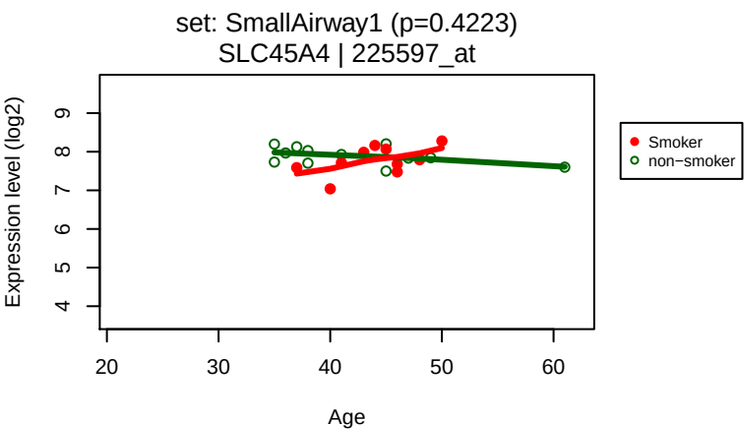
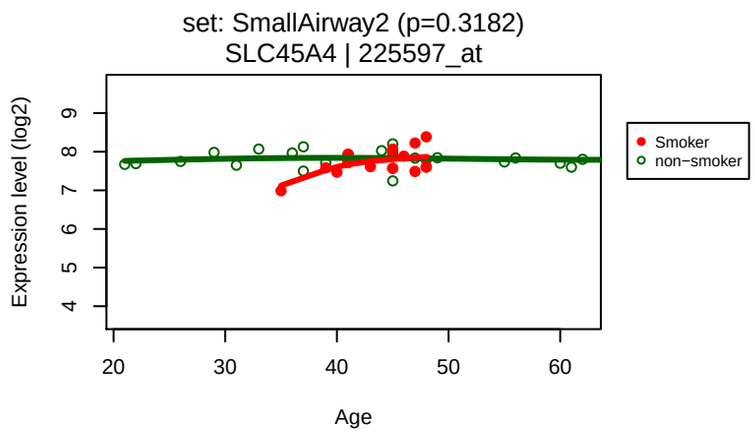
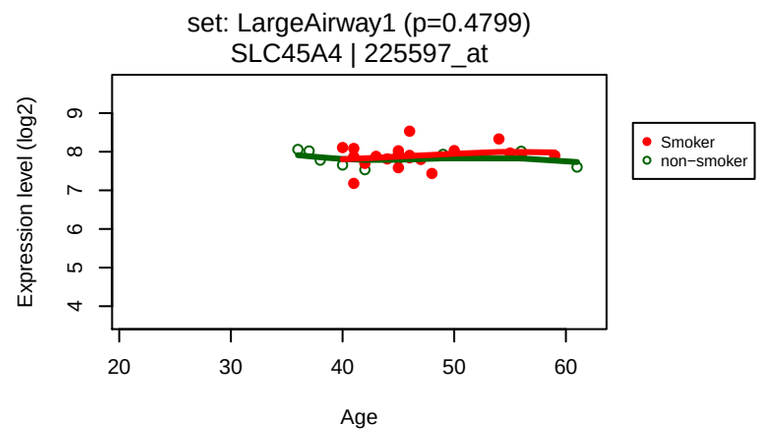

Gene expression of smokers and nonsmokers
gene: SLC45A4 | 225597_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4223
p=0.3182
p=0.4708
p=0.4799
p=0.1604
p=1e−04
p=0.7844
p=0.0567
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

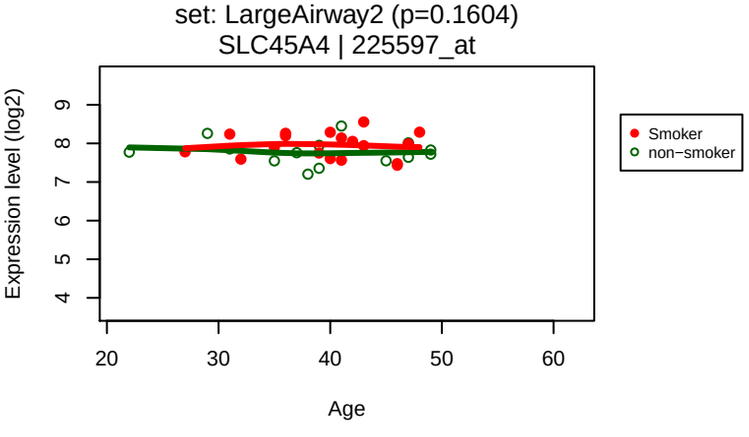

Gene expression ratio (smokers/nonsmokers)
gene: SLC45A4 | 225597_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4223
p=0.3182
p=0.4708
p=0.4799
p=0.1604
p=1e−04
p=0.7844
p=0.0567
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

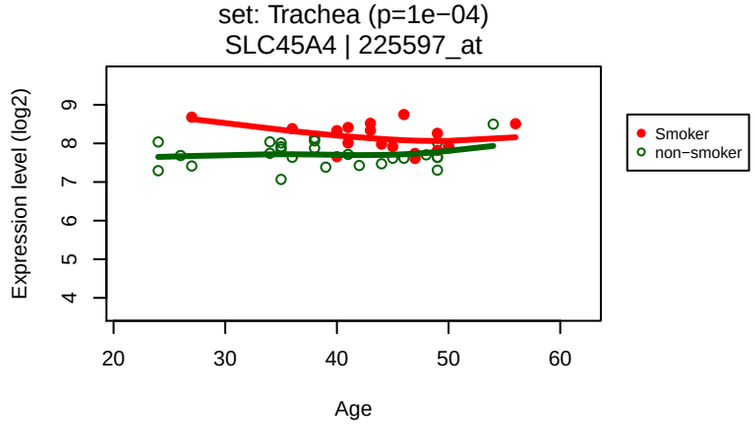

Gene expression ratio in Males
gene: SLC45A4 | 225597_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5199
p=0.0059
p=1e−04
p=0.0238
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC45A4 | 225597_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.273
p=0.4278
p=0.5211
p=0.8845
SmallAirway2
LargeAirway2
Trachea
Macrophage2

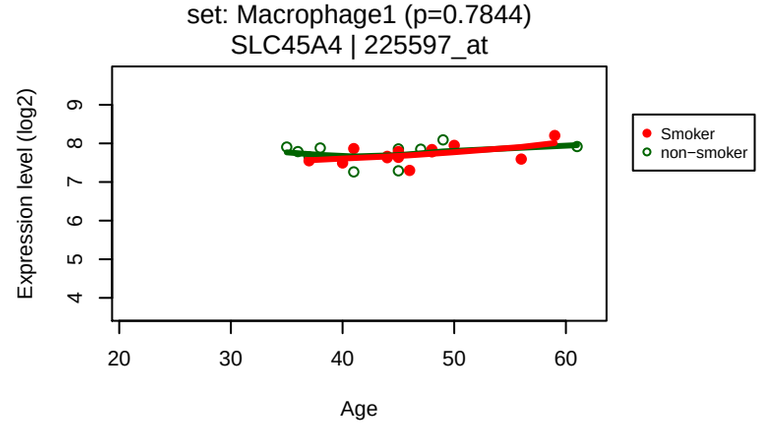

set: SmallAirway1 (p=0.4223)
SLC45A4 | 225597_at
Expression level (log2)
Age
Smoker
non−smoker

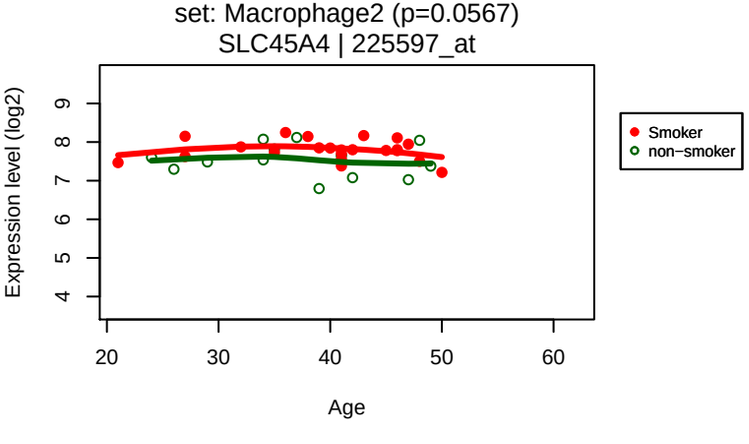

set: SmallAirway2 (p=0.3182)
SLC45A4 | 225597_at
Expression level (log2)
Age
Smoker
non−smoker

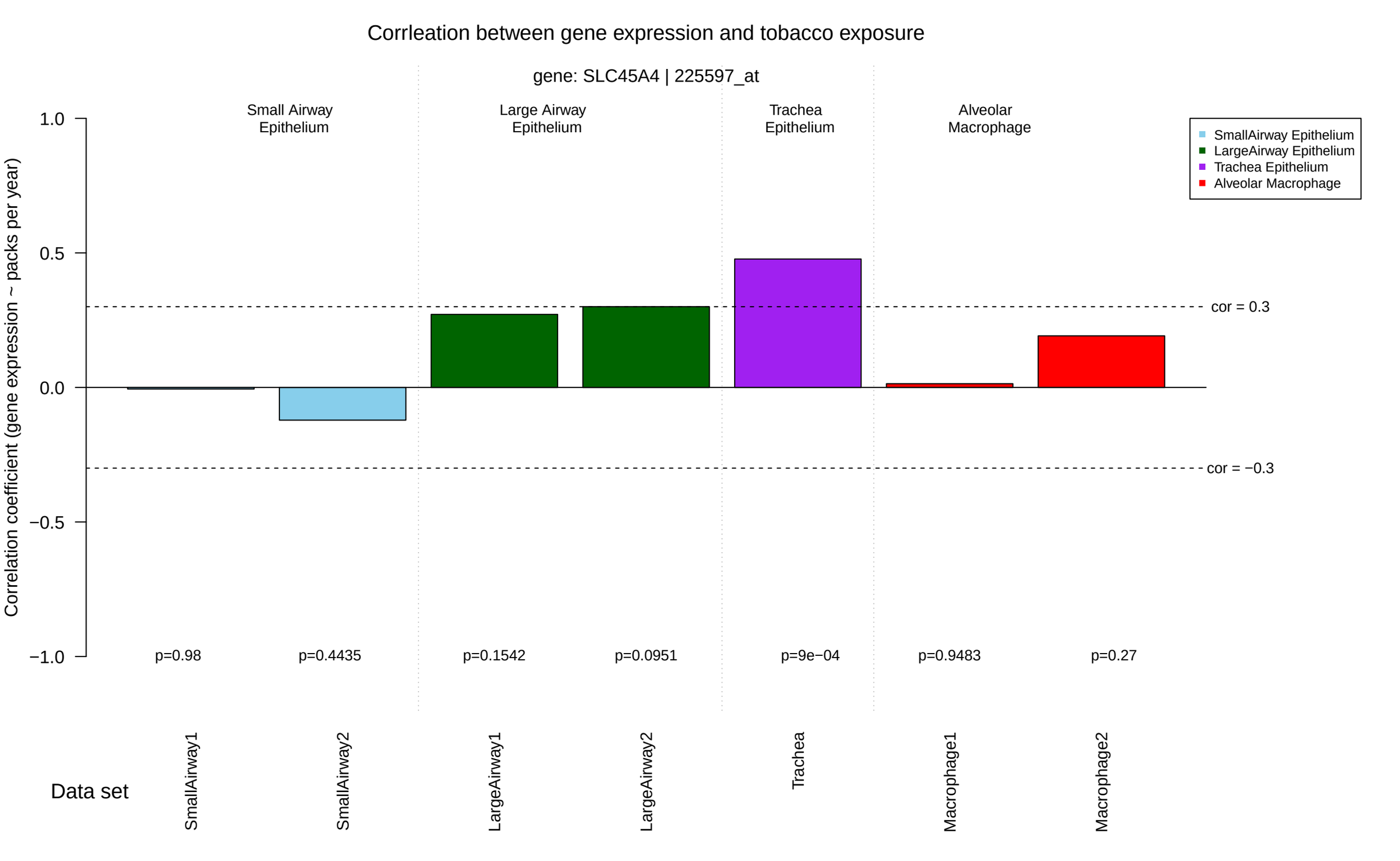

set: LargeAirway1 (p=0.4799)
SLC45A4 | 225597_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1604)
SLC45A4 | 225597_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
SLC45A4 | 225597_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7844)
SLC45A4 | 225597_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0567)
SLC45A4 | 225597_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC45A4 | 225597_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.98
p=0.4435
p=0.1542
p=0.0951
p=9e−04
p=0.9483
p=0.27
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2